Supplemental Material for “Real-Time Robust HOCBF Safety Filtering for Supercritical Power Plant Control under Model Mismatch”

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Supplemental Material

Reference numbers in this Supplemental Material follow the numbering of the main manuscript.

S1 Generic-GP and Out-of-Distribution Deployment

The main manuscript (Section 4) reports that scenario-specific GP training achieves 0% CBF violation under nominal operation and the six static perturbation deployments. This supplement provides the full OOD analysis: Table S1 reports CBF violation and QP intervention rates under three GP training strategies—scenario-specific, generic mixed (trained on Nominal/S1/S2/S4), and mismatched OOD (S1→S3, S3→S1)—using Φ-scaled rollout, $\kappa_\epsilon = 1.0$, and 5 seeds × 500 evaluation steps.

Three observations follow. (1) Scenario-specific GP is uniformly safe but with near-saturated QP intervention under perturbation. (2) Generic mixed GP fails under all perturbed deployments (82.7–99.7% violation) because it averages over training scenarios, producing σ_{GP} too small to trigger the safety filter—a silent calibration failure. (3) OOD

Table S1: Generic-GP and OOD deployment — CBF violation rate (%, mean $\pm$ std over 5 seeds $\times$ 500 steps).

Deployment	Strategy	CBF (%)	QP (%)
Nominal	scenario-specific	**0.0 ± 0.0**	0.2
	generic mixed	**0.0 ± 0.1**	43.0
S1: Heat	scenario-specific	**0.0 ± 0.0**	100.0
	generic mixed	92.2 ± 4.6	7.9
	OOD (S3→S1)	99.9 ± 0.1	99.8
S2: Pressure	scenario-specific	**0.0 ± 0.0**	99.8
	generic mixed	96.4 ± 0.6	3.4
S3: Coupled	scenario-specific	**0.0 ± 0.0**	99.2
	generic mixed	82.7 ± 3.9	18.6
	OOD (S1→S3)	**0.0 ± 0.0**	100.0
S4: Nonlinear	scenario-specific	**0.0 ± 0.0**	99.6
	generic mixed	98.2 ± 0.3	1.1
S5: Valve	scenario-specific	**0.0 ± 0.0**	99.6
	generic mixed	98.8 ± 0.3	0.2
S6: Fuel	scenario-specific	**0.0 ± 0.0**	99.9
	generic mixed	99.7 ± 0.2	0.2

asymmetry: S1→S3 achieves 0% violation (large σ_{GP} inflates $\epsilon(x)$) while S3→S1 reaches 99.9% violation (low σ_{GP} despite biased mean)—σ_{GP} measures spatial distance, not functional distance between perturbation structures.

S2 Perturbation Magnitude Sweep

Table S2 reports the perturbation magnitude sweep on S1:Heat (3 seeds per magnitude, Φ-scaled rollout, $\kappa_\epsilon = 1.0$). The QP intervention rate transitions smoothly from selective (3.5% at Mag10) to saturated (100% at Mag75), with 0% CBF violation through Mag75. At Mag100, the perturbation exceeds the GP's modeling capacity and violation reaches 99.8%.

Table S2: Perturbation magnitude sweep — CBF violation and QP intervention vs. enthalpy perturbation magnitude (S1:Heat structure, 3 seeds $\times$ 500 steps).

Magnitude	Δf_h (kJ/kg)	CBF%	QP%
Mag10	-10	0.0 ± 0.0	3.5
Moderate	-15	0.0 ± 0.0	9.8
Mag25	-25	0.0 ± 0.0	49.4
Mag50	-50	0.0 ± 0.0	99.9
Mag75	-75	0.0 ± 0.0	100.0
Mag100	-100	99.8 ± 0.0	100.0

S3 κ_ϵ Sensitivity Sweep

Table S3 reports the κ_ϵ sweep on S1:Heat and S3:Coupled under Φ-scaled nonlinear dynamics (3 seeds). All $\kappa_\epsilon \in [0.3, 1.0]$ achieve 0% CBF violation; Theorem 1 formally certifies only $\kappa_\epsilon = 1$. Reward improves monotonically with κ_ϵ and chatter shifts by < 1.0 percentage point.

Table S3: κ_ϵ sensitivity sweep (3 seeds, mean $\pm$ std).

κ_ϵ	Scenario	CBF %	Pwr %	Reward	Chatter %
0.30	S1	0.00 ± 0.00	0.00	-10919.9 ± 7.6	99.67
	S3	0.00 ± 0.00	0.00	-8089.5 ± 9.7	97.00
0.50	S1	0.00 ± 0.00	0.00	-10909.0 ± 7.8	99.67
	S3	0.00 ± 0.00	0.00	-8079.2 ± 9.7	97.40
0.70	S1	0.00 ± 0.00	0.00	-10898.3 ± 7.9	99.67
	S3	0.00 ± 0.00	0.00	-8068.9 ± 9.8	97.53
1.00	S1	$\mathbf{0.00 \pm 0.00}$	0.00	-10882.6 ± 8.3	99.67
	S3	$\mathbf{0.00 \pm 0.00}$	0.00	-8053.8 ± 9.9	97.93

S4 Training Mode Ablation

Table S4 compares three training modes under S1:Heat. All achieve 0% CBF violation at deployment; decoupled mode requires 12% less training time; coupled-differentiable is computationally infeasible (>30 min/seed).

Table S4: Training Mode Ablation under S1: Heat (mean $\pm$ std over 3 seeds, 30 eval episodes $\times$ 300 steps).

Mode	CBF%	Reward	Train Time	Solve (ms)
Decoupled	0.0	-4576.2 ± 1.8	12.5 ± 0.7 min	$\sim$25
Coupled-nondiff.	0.0	-4580.1 ± 2.1	14.2 ± 0.8 min	$\sim$25
Coupled-diff.	0.0	-4582.4 ± 2.5	> 30 min (no conv.)	$\sim$25

S5 Computation Time Analysis

Per-step filter latency is a critical metric for real-time industrial deployment. Figure S1 compares the per-step computation time of the Robust HOCBF filter (PPO forward pass + GP inference + QP solve, GPU, JIT-compiled) against the implemented SLSQP-based NMPC baseline (CPU, N=5 horizon). The filter provides per-step rather than horizon-based safety evaluation.

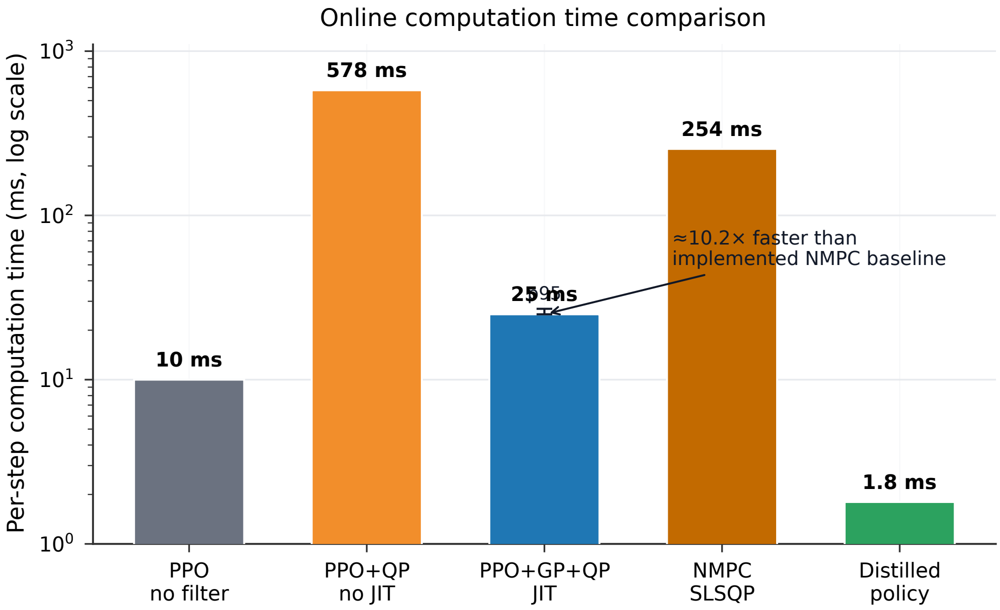

Figure S1: Per-step computation time comparison (log scale). PPO+GP+QP with JIT compilation (GPU): ∼25 ms (p95: 27 ms); NMPC (CPU, SLSQP, N=5 horizon): 254 ms average.

S6 GP–CBF Structural Comparison

Table S5 (moved from Section 2) summarizes the key structural distinctions between GP-augmented safety methods.

Table S5: GP-augmented safety methods compared on three structural axes.

	Structural axis		
Method	Uncertainty propagation depth	Per-constraint differentiation	GP-UCB guarantee type
Cheng et al. (2019)	Top-level CBF only	No (single margin)	Per-step CBF (m=1)
Hewing et al. (2020)	MPC prediction horizon (rollout)	Partial (per chance constraint, rollout-aggregated)	Chance-constrained over horizon
Ours (Robust HOCBF)	Full HOCBF ψ-chain (all m levels)	Yes (per constraint, by relative degree)	Per-step forward invariance ($m \geq 1$)

S7 PPO-Lagrangian Baseline Results

PPO-Lagrangian uses dual descent on a cost constraint to penalize unsafe behavior during training, providing a constrained-RL baseline. Table S6 reports its CBF violation rates on the available Phase 4 conditions. PPO-Lagrangian achieves substantially lower violation rates than the nominal PPO-HOCBF ($>$ 95% under perturbation) but does not achieve the zero-violation performance of PPO-RHOCBF (0% across all static scenarios). The Lagrangian penalty mechanism provides partial safety through reward shaping but cannot guarantee constraint satisfaction under model mismatch—the dual variable tracks average constraint cost, not worst-case per-step forward invariance.

Table S6: PPO-Lagrangian CBF Violation Rate (%) under Φ-Scaled Nonlinear Dynamics (mean ± std over 5 seeds, 50 eval episodes × 500 steps).

Method	Nom.	S1: Heat	S2: Press.	S3: Coupled	S4: Nonlin.
PPO-Lagrangian	0.14 ± 0.12	1.27 ± 0.93	1.04 ± 0.39	0.47 ± 0.15	0.75 ± 0.31
PPO-HOCBF	0.0	100.0	99.9	99.9	99.9
PPO-RHOCBF	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**

PPO-Lagrangian with 200 training episodes × 200 steps per episode converges to dual-variable equilibrium but the dual descent mechanism does not provide per-step forward invariance certification. S5 (Valve) and S6 (Fuel) are 5th-order scenarios not included in the Phase 4 experimental configuration; PPO-Lagrangian was not evaluated on these conditions.

S8 Load-Following Performance

Load following tests the control system's ability to track an AGC (Automatic Generation Control) power schedule while maintaining safety constraints. The schedule ramps between 750 MW and 1000 MW at 5 MW/min with 20 MW regulation amplitude and 300 s regulation period. Table S7 reports CBF violation rates under this dynamic reference trajectory.

Table S7: CBF Violation Rate (%) under Load-Following AGC Schedule (mean ± std over 5 seeds, 50 episodes × 500 steps).

Method	CBF Viol.%
PPO (no safety)	0.05 ± 0.09
PPO-Lagrangian	0.05 ± 0.09
PPO-CBF (m=1, incorrect)	0.70 ± 0.36
PPO-HOCBF (correct m)	0.26 ± 0.12
PPO-RHOCBF	$\mathbf{0.11 \pm 0.05}$
RoCBF-Net	$\mathbf{0.14 \pm 0.19}$
NMPC (N=5, offset-free)	0.00 ± 0.00

Under load following, the nominal dynamics are exact ($\Delta f = 0$) and the CBF constraints are only challenged by the reference trajectory changes, not by model mismatch. NMPC achieves 0% violation because its offset-free disturbance estimator converges quickly on the exact model. PPO-RHOCBF and RoCBF-Net maintain near-zero violation with the fixed scenario-specific GP providing a small robustness margin. The low violation rates across all methods confirm that load following is a less challenging condition than persistent model mismatch.

S9 Triple-Integrator Validation ($m = 3$)

The compositional robustness margin $\epsilon(x)$ is derived for general relative degree m (Theorem 1 of the main paper), but the CCS constraints have $m \leq 2$. This appendix validates the $m \geq 3$ generalization on a triple integrator system ($n = 3$, $m = 3$): $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, $\dot{x}_3 = u$, with a circular keep-out zone $h(x) = (x_1 - c)^2 - r^2$ of relative degree $m = 3$. We evaluate four perturbation types: additive damping ($\Delta f = -\zeta[0, 0, x_2]^\top$, $\zeta = 0.5$), periodic ($\Delta f = A \sin(\omega t)[0, 0, 1]^\top$), coupled (damping + periodic), and nonlinear state-dependent ($\Delta f = -\eta[0, 0, x_1 x_2]^\top$). Each method is evaluated over 50 episodes of 500 steps.

Sigma-chain validation A key property of the recursive σ-chain is that uncertainty grows monotonically with the ψ-chain level: $\sigma_1 < \sigma_2 < \sigma_3$. Table S8 validates this across 120 sampled states and all four perturbation scenarios. The ordering $\sigma_1 < \sigma_2 < \sigma_3$ holds for 100% of sampled states, with growth ratios $\sigma_2/\sigma_1 \approx 2.6$–$2.9$ and $\sigma_3/\sigma_2 \approx 2.0$–$2.2$, consistent with the recursive structure of the σ-chain (Eq. 7 of the main manuscript).

Table S8: Sigma-chain ordering validation ($m = 3$): mean σ_i values and growth ratios across 120 sampled states

Scenario	$\bar{\sigma}_1$	$\bar{\sigma}_2$	$\bar{\sigma}_3$	$\sigma_1 < \sigma_2 < \sigma_3$	σ_3/σ_1
Damping	0.077	0.207	0.437	100%	5.7
Periodic	0.101	0.254	0.514	100%	5.1
Coupled	0.101	0.254	0.514	100%	5.1
Nonlinear	0.077	0.219	0.474	100%	6.1

Safety evaluation Table S9 reports the safety results. The critical result is under the *nonlinear* state-dependent perturbation: HOCBF(m=3) suffers 13.3% actual constraint violation and 41.5% CBF condition violation, while RHOCBF(m=3) achieves 0% actual constraint violation with only 3.5% CBF condition violation. The robustness margin $\epsilon(x)$ reduces actual violation from 13.3% to 0%.

The 3.5% CBF condition violation under the nonlinear scenario reflects that the large $\bar{\epsilon} = 20.1$ occasionally renders the QP infeasible, in which case the raw policy action may violate the CBF condition. Crucially, the system remains a safe distance from the actual constraint boundary, achieving 0% actual violation—the compositional $\epsilon(x)$ provides practical safety even when formal forward invariance cannot be guaranteed due to QP infeasibility.

Compositional vs. constant ϵ The nonlinear scenario provides a setting where $\epsilon(x)$ varies substantially ($\mu_\epsilon = 0.93$, $\max_\epsilon = 3.39$, std/$\mu = 0.70$). Table S10 reports results

Table S9: Triple Integrator Validation ($m = 3$): Constraint Violation Rate (%) under four perturbation types (50 episodes $\times$ 500 steps). "Actual" = entering keep-out zone ($h < 0$); "CBF" = CBF condition violation ($\psi_m < 0$).

Scenario	Method	Actual Viol.	CBF Viol.
Damping	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Periodic	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Coupled	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Nonlinear	HOCBF(m=3)	13.25	41.45
	RHOCBF(m=3)	**0.00**	3.50

across five seeds. All ϵ configurations achieve 0% actual violation, but CBF condition violation rates differ: compositional $\epsilon(x)$ (3.30%) outperforms ϵ_0^{mean} (4.70%) and underperforms ϵ_0^{max} (1.50%)—the compositional $\epsilon(x)$ balances conservatism by being tighter in well-observed states while maintaining protection at the boundary.

Table S10: Triple Integrator ϵ Ablation under Nonlinear Perturbation ($m = 3$, 5 seeds $\times$ 50 episodes $\times$ 500 steps). Mean ϵ is the total per-step robustness margin aggregated over all evaluation steps. The compositional value reflects the accumulated σ-chain contribution entering the $m = 3$ HOCBF inequality, whereas ϵ_0^{mean} and ϵ_0^{max} are constant baseline margins. Results: mean $\pm$ std over 5 seeds.

ϵ Configuration	Actual Viol. (%)	CBF Viol. (%)	Mean ϵ
Compositional $\epsilon(x)$	0.00 ± 0.00	3.30 ± 0.24	21.6
ϵ_0^{mean}	0.00 ± 0.00	4.70 ± 0.24	0.93
ϵ_0^{max}	0.00 ± 0.00	1.50 ± 0.00	3.39
No ϵ ($\epsilon = 0$)	13.43 ± 0.12	41.23 ± 0.07	0

Sparse GP coverage and task performance A position constraint $h(x) = x_{\text{lim}} - x_1$ with constant gradient $|\partial h/\partial x_1| = 1$ combined with sparse GP ($n = 50$ samples confined to $x_1 \in [-0.3, 0.5]$, constraint at $x_{\text{lim}} = 1.2$) creates $17\times$ $\epsilon(x)$ variation. Table S11 shows that compositional $\epsilon(x)$ enables task completion that constant ϵ_0 prevents: $\bar{x}_1 = 0.37$ (31% of distance to constraint) with 0% violation vs. $\bar{x}_1 = -0.38$ and -1.33 for constant margins. Without ϵ, the state reaches $\bar{x}_1 = 0.59$ but with 12.2% violation.

Table S11: Sparse GP Ablation: Position Constraint with Partial GP Coverage ($m = 3$, nonlinear perturbation, 5 seeds $\times$ 10 episodes $\times$ 200 steps). GP trained on n=50 samples in $x_1 \in [-0.3, 0.5]$; constraint at $x_{\text{lim}} = 1.2$. Results: mean $\pm$ std over 5 seeds.

ϵ Config.	Viol. (%)	CBF Viol. (%)	QP Infeas. (%)	$\bar{x}_1$	$x_1^{\max}$	Mean ϵ
Compositional $\epsilon(x)$	0.00	0.00	0.0	**0.37**	**0.74**	0.15
ϵ_0^{mean}	0.00	0.00	0.0	-0.38	0.00	1.20
$\epsilon_0^{\max}$	0.00	0.00	0.2	-1.33	0.00	2.72
No ϵ ($\epsilon = 0$)	12.2 ± 0.0	56.8 ± 0.7	0.0	0.59	1.35	0

S10 Proofs of Theoretical Results

S10.1 Residual Gradient Bound

Lemma 1 (Residual gradient bound). *Let $\delta_i = \psi_i - \hat{\psi}_i^0$ with $\delta_0 \equiv 0$. Under the smoothness and gradient boundedness conditions stated in Theorem 1 of the main manuscript and smoothness of $\hat{f}$ on the compact operating region $\mathcal{X}_{\text{op}}$, $\|\nabla_x \delta_i(x)\| \leq G_\delta^{(i)}$ for all $x \in \mathcal{X}_{\text{op}}$, with*

$$\begin{aligned}
&G_\delta^{(0)} = 0, \\
&G_\delta^{(i)} = H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f + (L_{\hat{f},i} + k_i) G_\delta^{(i-1)} + H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f
\end{aligned} \tag{S1}$$

where $H_\psi^{(i-1)} = \sup_x \|\nabla_x^2 \hat{\psi}_{i-1}^0\|_{\text{op}}$, $G_\psi^{(i-1)} = \sup_x \|\nabla_x \hat{\psi}_{i-1}^0\|$, and G_f is the gradient bound from the gradient boundedness condition of Theorem 1 of the main manuscript.

Proof. Differentiate δ_i term by term: $\delta_i = L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i)\delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1}$.

(i) $\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0) = \nabla_x^2 \hat{\psi}_{i-1}^0 \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \hat{\psi}_{i-1}^0$. By Cauchy-Schwarz and the gradient boundedness condition of Theorem 1 of the main manuscript: $\|\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0)\| \leq H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f$.

(ii) $\nabla_x ((L_{\hat{f}} + k_i)\delta_{i-1})$ contributes $(L_{\hat{f},i} + k_i) G_\delta^{(i-1)}$.

(iii) $\nabla_x (L_{\Delta \hat{f}} \delta_{i-1}) = \nabla_x^2 \delta_{i-1} \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \delta_{i-1}$, bounded by $H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f$.

Summing yields Eq. (S1). Under the gradient boundedness condition of Theorem 1 of the main manuscript, $G_f = L_{\hat{f}} \|\Delta \hat{f}\|_\infty = \mathcal{O}(\|\Delta \hat{f}\|)$. By induction, $G_\delta^{(i)} = \mathcal{O}(\|\Delta \hat{f}\|)$ at every level, so $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \|\Delta \hat{f}\| = \mathcal{O}(\|\Delta \hat{f}\|^2) = \mathcal{O}(\rho^2)$. □

S10.2 Full Proof of Theorem 1

Proof of Theorem 1. **Step 1 (GP calibration).** By the RKHS boundedness and GP calibration conditions stated in Theorem 1 of the main manuscript and the GP-UCB concentration inequality of Srinivas et al. [4], $|\Delta f_j(x) - \mu_{\text{GP},j}(x)| \leq \beta \cdot \sigma_{\text{GP},j}(x)$ for all

$x \in \mathcal{X}$ holds with probability $\geq 1-\delta/n$ per dimension. By union bound over $j = 1, \ldots, n$, this holds simultaneously with probability $\geq 1 - \delta$. Condition on this event.

Step 2 (Compositional bound). We show $|\delta_i(x)| \leq \sigma_i(x)$ by induction on i.

- *Base case ($i = 1$):* $\delta_1 = \sum_j (\partial h/\partial x_j)\Delta\hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \leq \beta \sum_j |\partial h/\partial x_j| \sigma_{\mathrm{GP},j} = \sigma_1$.
- *Inductive step ($i \geq 2$):* Assume $|\delta_{i-1}| \leq \sigma_{i-1}$. From Eq. (5) of the main text: $|\delta_i| \leq |L_{\Delta\hat{f}}\hat{\psi}^0_{i-1}| + (|L_{\hat{f}}|_{\mathrm{op}} + k_i)|\delta_{i-1}| + |L_{\Delta\hat{f}}\delta_{i-1}|$. Term 1 bounded identically to base case; Term 2 bounded by induction hypothesis; Term 3 bounded using Lemma S1: $|L_{\Delta\hat{f}}\delta_{i-1}| \leq G_\delta^{(i-1)}\beta\|\sigma_{\mathrm{GP}}\|_2 = \sigma^{(i)}_{\mathrm{cross}}$. Summing: $|\delta_i| \leq \sigma_i$.

The cross term is retained at full weight. Under the gradient boundedness condition of Theorem 1 of the main manuscript, it is formally $\mathcal{O}(\rho^2)$.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound Eq. (S2) yields $|\Delta(x,u)| \leq \sigma_m(x) + \sum_{j=1}^{m-1} c_j\sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) = \sigma_{\mathrm{total}}(x)$. For $\kappa_\epsilon = 1$, $\epsilon(x) = \sigma_{\mathrm{total}}(x)$, hence $|\Delta(x,u)| \leq \epsilon(x)$.

Step 4 (Forward invariance). For u^* satisfying $A_0u^* \leq b_0 - \epsilon$: $\psi_m^{\mathrm{true}}(x,u^*) = \hat{\psi}^0_m(x,u^*) + \Delta(x,u^*) \geq \epsilon(x) - |\Delta(x,u^*)| \geq 0$. By Theorem 2 of Xiao et al. [2], the safe set $\mathcal{C}$ is forward invariant. □

S10.3 Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\mathrm{ctrl}}(x) = \beta \sum_j \left| \frac{\partial(L_g L_{\hat{f}}^{m-1}h)}{\partial x_j} \right| \sigma_{\mathrm{GP},j}(x) \cdot u_{\max} \tag{S2}$$

where $u_{\max} = \sup_{u\in\mathcal{U}} \|u\|$. On the CCS, $\sigma_{\mathrm{ctrl}}/\sigma_{\mathrm{total}} < 0.15$ for both constraints.

References